

Probability and Statistics (MT-2005)

Homework # 7

Statistical Properties of Continuous Probability Distributions: Normal Distribution

Questions

1. Transformation of a Normal Distribution

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Define a new variable $Y = aX + b$, where $a \neq 0$ and $b \in \mathbb{R}$.

- (a) Prove that Y is also normally distributed. Specify its mean and variance.

Hint: Use the moment generating function (MGF) or apply the PDF transformation technique.

- (b) If $a = -2$, $b = 5$, and $X \sim \mathcal{N}(3, 4)$, find the distribution of Y .

- (c) Prove that the sum of two independent normal random variables is also normally distributed. i.e., if $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ then prove that $X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

2. Median, Mode, and Quartiles of the Normal Distribution

Let $X \sim \mathcal{N}(\mu, \sigma^2)$.

- (a) Find the median of X .

Hint: Recall that the median m satisfies the equation:

$$\int_{-\infty}^m \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = \frac{1}{2}$$

Try making the substitution $z = \frac{x-\mu}{\sigma}$, which transforms the integral into one over the standard normal distribution. Then, use the symmetry of the standard normal curve.

- (b) Find the mode of X .

Hint: Use the first derivative of the PDF to locate the maximum point.

- (c) Find the first quartile Q_1 of X .

Hint: Use the standard normal table to find the z -value corresponding to the 25th percentile.

3. Maximum Point and Inflection Points of the Normal Curve

Let $f(x)$ be the PDF of a normal distribution:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- (a) Find the x -value at which $f(x)$ is maximum.

Hint: Differentiate $f(x)$ and find where $f'(x) = 0$.

- (b) Determine the points of inflection of the curve.

Hint: Compute the second derivative $f''(x)$ and solve $f''(x) = 0$.